

ZDVINSK, Russian Federation

WMO: 297120

Lat: **54.717N**Lon: **78.667E**Elev: **113**StdP: **99.97**Time Zone: **7.00 (E07)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-35.9	-32.4	-38.7	0.1	-35.8	-35.3	0.1	-32.3	12.3	-10.1	10.4	-9.6	1.6	200	0.645

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.4	29.8	18.8	28.3	18.4	26.6	17.8	20.7	26.6	19.7	25.6	18.9	24.5	3.9	150

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.7	13.7	23.0	17.7	12.9	22.4	16.7	12.1	21.6	60.3	26.6	57.1	25.9	54.4	24.6	25.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 10.6	9.2	8.2	-38.2	32.9	4.3	2.0	-41.3	34.3	-43.8	35.5	-46.3	36.6	-49.4	38.0		
(5)			-38.2	22.3	4.2	1.5	-41.3	23.4	-43.7	24.2	-46.1	25.1	-49.1	26.1		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	1.8	-18.5	-15.9	-7.4	4.1	12.1	17.5	19.3	16.3	10.8	3.2	-6.7	-14.5		
	DBStd	14.57	8.27	7.41	6.42	5.84	5.49	4.08	3.23	3.87	4.68	5.06	7.68	8.02		
	HDD10.0	3915	884	726	539	193	43	3	0	2	44	218	501	761		
	HDD18.3	6176	1143	959	798	428	203	63	27	86	230	469	751	1019		
	CDD10.0	914	0	0	0	15	109	226	289	198	69	7	0	0		
	CDD18.3	134	0	0	0	0	10	36	59	24	4	0	0	0		
	CDH23.3	1356	0	0	0	4	137	366	522	268	58	1	0	0		
	CDH26.7	341	0	0	0	0	29	89	142	67	14	0	0	0		
(14) Wind	WSAvg	3.6	3.5	3.6	4.1	4.0	4.1	3.3	2.8	3.0	3.2	3.6	3.9	4.0		
(15) Precipitation	PrecAvg	359	17	12	15	17	26	50	55	47	33	33	28	25		
	PrecMax	483	34	23	40	43	81	102	146	110	74	75	56	44		
	PrecMin	208	2	1	1	3	5	18	9	7	4	10	10	6		
	PrecStd	69	9	6	11	9	16	23	30	25	17	19	12	10		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	0.5	0.8	6.2	23.2	29.5	31.0	32.4	31.2	29.1	20.0	8.2	1.4		
		MCWB	-0.2	0.0	3.2	13.2	17.1	18.6	20.2	19.0	17.0	11.3	6.2	0.4		
	2%	DB	-1.9	-0.9	3.6	19.4	26.8	29.2	29.9	28.4	24.8	16.4	5.9	0.2		
		MCWB	-2.5	-1.6	1.6	12.0	15.7	18.3	19.6	18.7	15.0	9.9	4.1	-0.3		
	5%	DB	-4.1	-2.8	2.2	16.2	24.6	27.3	28.2	26.3	21.6	13.3	3.7	-1.6		
		MCWB	-4.7	-3.5	0.8	10.1	15.2	17.9	19.3	17.6	13.6	8.2	2.5	-2.1		
	10%	DB	-6.1	-5.0	1.0	13.3	21.9	25.3	26.3	24.2	18.9	10.5	2.0	-3.8		
		MCWB	-6.6	-5.7	-0.2	8.4	13.4	17.2	18.7	16.8	12.5	6.8	1.0	-4.4		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-0.1	0.3	3.2	14.4	18.7	21.2	22.1	21.4	18.1	12.3	6.8	0.6		
		MCDB	0.3	0.7	5.1	21.1	27.1	27.8	28.4	26.5	25.5	17.8	7.8	1.0		
	2%	WB	-2.5	-1.6	2.0	12.3	17.0	19.8	21.1	19.7	15.6	10.3	4.3	-0.3		
		MCDB	-2.0	-1.0	3.3	18.1	25.0	26.5	27.0	26.0	23.0	15.5	5.6	0.2		
	5%	WB	-4.6	-3.5	1.0	10.4	15.6	18.8	20.1	18.7	14.3	8.6	2.6	-2.1		
		MCDB	-4.1	-2.9	2.1	15.8	23.1	24.9	25.7	24.5	20.5	12.6	3.7	-1.6		
	10%	WB	-6.7	-5.6	-0.1	8.5	14.2	17.9	19.4	17.5	13.0	7.1	1.1	-4.3		
		MCDB	-6.2	-4.9	1.0	12.9	20.7	23.7	24.7	23.1	18.2	10.3	1.9	-3.8		
(35) Mean Daily Temperature Range	5% DB	MDBR	8.2	8.7	8.8	9.4	11.5	10.7	10.4	11.0	10.4	8.0	6.2	7.6		
		MCDBR	8.5	7.2	7.6	14.2	14.9	14.0	13.7	14.4	15.0	13.0	6.1	7.5		
	5% WB	MCWBR	8.3	7.1	6.4	8.1	6.9	5.8	5.4	6.2	7.0	7.4	4.9	7.4		
		MCWBR	8.3	7.2	6.8	13.3	13.5	11.9	11.2	12.5	13.5	11.7	5.9	7.7		
(40) Clear-Sky Solar Irradiance	taub		0.291	0.322	0.378	0.406	0.406	0.409	0.436	0.419	0.369	0.353	0.325	0.293		
		taud	2.023	1.981	1.920	2.128	2.215	2.287	2.209	2.229	2.322	2.233	2.077	1.975		
	Ebn at Noon		603	722	764	812	837	838	804	793	789	695	553	493		
		Edh at Noon	71	106	144	135	131	124	131	121	97	83	68	60		
(44) All-Sky Solar Radiation	RadAvg		0.99	2.04	3.53	4.68	5.62	6.01	5.46	4.55	3.17	1.62	0.77	0.64		
	RadStd		0.08	0.11	0.23	0.34	0.48	0.51	0.37	0.37	0.32	0.18	0.12	0.08		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (13 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page